**BRAC University**

Department of Mathematics and Natural Sciences


LECTURE ON

**Real Analysis (MAT221)**

# Convergence Tests of Infinite Series

## Ratio Test, Root Test, P-series Test, Alternating Test

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CONDUCTED BY


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# Infinite Series

## Infinite Series

Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of real numbers. The infinite series generated by this sequence is the formal expression

$$\sum_{n=1}^{\infty} a_n.$$

# Convergence of Infinite Series

## Convergence of Infinite Series

We define the corresponding sequence of partial sums $(s_m)$ by

$$s_m = a_1 + a_2 + a_3 + \cdots + a_m,$$

and say that the series $\sum_{n=1}^{\infty} a_n$ converges to $S$ if the sequence $(s_m)$ converges to $S$. In this case, we write

$$\sum_{n=1}^{\infty} a_n = S.$$

If the sequence $\{S_N\}$ does not converge, the series is called divergent.

## Convergence of Geometric Series

### Theorem

The geometric series

$$\sum_{k=0}^{\infty} a + ar + ar^2 + ar^3 + \cdots$$

converges if and only if $|r| < 1$.

## Comparison Test

### Theorem

Let $\{a_n\}$ and $\{b_n\}$ be sequences of non-negative real numbers such that

$$0 \leq a_n \leq b_n \qquad \text{for all } n.$$

Then:

(i) If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ also converges.

(ii) If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ also diverges.

# Problems on Comparison Test

## Problem

Determine whether the series

$$\sum_{n=1}^{\infty} \frac{1}{n^2+3}$$

converges or diverges.

## Problem

Determine whether the series

$$\sum_{n=1}^{\infty} \frac{1}{n^2 - 3}$$

converges or diverges.

**Problem**

Test the convergence of
$$\sum_{n=1}^{\infty} \frac{5n}{n^3+1}.$$

**Problem**

Discuss the convergence of

$$\sum_{n=1}^{\infty} \frac{7}{\sqrt{n^3 + 4n}}.$$

## Problem

Determine whether

$$\sum_{n=1}^{\infty} \frac{4n+1}{n^2+n}$$

converges or diverges.

## Problem

Test the convergence of

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}+5}.$$

## p-series Test

### Theorem

Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{n^p},$$

where $p$ is a real number.

(i) If $p > 1$, then the series converges.

(ii) If $p \leq 1$, then the series diverges.

## Problems on p-Series Test

### Problem

Determine whether the series

$$\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$$

converges or diverges.

**Problem**

Investigate the convergence of

$$\sum_{n=1}^{\infty} \frac{1}{n^{5/4}}.$$

**Problem**

Determine whether

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

converges.

**Problem**

Test the convergence of
$$\sum_{n=1}^{\infty} \frac{1}{n^{0.9}}.$$

# Limit Comparison Test

## Theorem

Let $\sum a_n$ and $\sum b_n$ be two positive–term series.

(i) If

$$\lim_{n\to\infty} \frac{a_n}{b_n} = L \qquad \text{with } 0 < L < \infty,$$

then the series $\sum a_n$ and $\sum b_n$ either both converge or both diverge.

(ii) If

$$\lim_{n\to\infty} \frac{a_n}{b_n} = 0 \quad \text{and } \sum b_n \text{ converges,}$$

then $\sum a_n$ also converges.

(iii) If

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \infty \quad \text{and } \sum b_n \text{ diverges,}$$

then $\sum a_n$ also diverges.

## Problems on Limit Comparison Test

### Problem

Test the convergence of

$$\sum_{n=1}^{\infty} \frac{3n^2+1}{5n^3+4}.$$

## Problem

Determine whether

$$\sum_{n=1}^{\infty} \frac{n+7}{n^2+9}$$

converges.

**Problem**

Discuss the convergence of

$$\sum_{n=1}^{\infty} \frac{4n}{\sqrt{n^3+1}}.$$

## Problem

Test whether

$$\sum_{n=1}^{\infty} \frac{2n^3 - n}{7n^3 + 9}$$

converges.

## Problem

Check the convergence of

$$\sum_{n=1}^{\infty} \frac{5}{n+n^2}.$$

## D'Alembert's Ratio Test

### Theorem

Let $\sum a_n$ be a series with positive terms, and suppose the limit

$$L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$$

exists (or is infinite). Then:

(i) If $L < 1$, the series $\sum a_n$ converges.

(ii) If $L > 1$ (including $L = \infty$), the series $\sum a_n$ diverges.

(iii) If $L = 1$, the test is inconclusive.

## Problems on D'Alembert's Ratio Test

### Problem

Test the convergence of

$$\sum_{n=1}^{\infty} \frac{3^n}{5^n}.$$

## Problem

Determine whether the series

$$\sum_{n=1}^{\infty} \frac{n!}{5^n}$$

converges.

## Problem

Test the convergence of

$$\sum_{n=1}^{\infty} \frac{2^n}{n!}.$$

## Problem

Investigate the convergence of

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n}.$$

**Problem**

Test whether the series

$$\sum_{n=1}^{\infty} \frac{4^n}{n^3}$$

converges.

# Cauchy's Root Test

## Theorem

Let $\sum a_n$ be a series with non-negative terms, and suppose the limit

$$L = \lim_{n\to\infty} \sqrt[n]{a_n}$$

exists (or is infinite). Then:

(i) If $L < 1$, the series $\sum a_n$ converges.

(ii) If $L > 1$ or $L = \infty$, the series $\sum a_n$ diverges.

(iii) If $L = 1$, the test is inconclusive.

## Problems on Cauchy's Root Test

### Problem

Test the convergence of

$$\sum_{n=1}^{\infty} \sqrt[n]{\frac{1}{3^n}}.$$

**Problem**

Determine whether

$$\sum_{n=1}^{\infty} \left(\frac{2}{5}\right)^n$$

converges using the root test.

## Problem

Discuss the convergence of

$$\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^n .$$

## Problem

Test the convergence of

$$\sum_{n=1}^{\infty} \left( \frac{n}{2n+1} \right)^n.$$

**Problem**

Determine whether the series

$$\sum_{n=1}^{\infty} \left(\frac{n^2}{3^n}\right)$$

converges.

## Root test is more general than Ratio test.

### Problem

Discuss the convergence of the series $\sum u_n$ where

$$u_n = \begin{cases} 2^{-n}, & \text{if } n \text{ is odd,} \\ 2^{-n+2}, & \text{if } n \text{ is even.} \end{cases}$$

# Cauchy Condensation Test

## Theorem

Let $\{a_n\}$ be a non-increasing sequence of positive real numbers. Then the series

$$\sum_{n=1}^{\infty} a_n$$

converges if and only if the series

$$\sum_{n=1}^{\infty} 2^n a_{2^n}$$

converges.

# Problems on Cauchy Condensation Test

## Problem

Test the convergence of

$$\sum_{n=1}^{\infty} \frac{1}{n \ln n}.$$

## Problem

Test the convergence of

$$\sum_{n=1}^{\infty} \frac{1}{n(\ln n)^2}.$$

**Problem**

Investigate the convergence of

$$\sum_{n=1}^{\infty} \frac{1}{n \ln(\ln n)}.$$

# Thank You!

We'd love your questions and feedback.


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## References


[1] Stephen Abbott, *Understanding Analysis*, 2nd Edition, Springer, 2015.

[2] Terence Tao, *Analysis I*, 3rd Edition, Texts and Readings in Mathematics, Hindustan Book Agency, 2016.